The Addressability Threshold and the Drop Rate: A Theory of Logistics Viability in Informal-Address Markets

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November 3, 2026

Abstract

Platform logistics firms in Kenya and comparable African markets have failed repeatedly despite abundant demand and substantial capital. This paper argues that these failures are not primarily failures of execution, pricing, or funding, but the consequence of a structural variable largely absent from the logistics literature: **addressability**, the degree to which a delivery destination can be located without human negotiation. We show that addressability is the **sole exogenous primitive** in the unit economics of last-mile delivery, operating through a single intermediate variable—the **drop rate** $R(a)$, the number of drops a vehicle can complete per unit time. Every downstream quantity—driver cost per drop, capacity per vehicle-hour, drops per trip, break-even volume, and the feasibility of a guaranteed service window—is monotone in $R(a)$. We derive a rate equation, an addressability-augmented cost function, a throughput constraint, and a critical addressability threshold a^* below which per-drop contribution is negative at any volume. We extend the model with a **Registry Proposition**, a **Ledger Proposition**, an **Aggregation Proposition**, and a **Risk Proposition**. The paper's central claim is conditional and testable: viability in informal-address markets requires holding either drop value or effective addressability above the threshold, and only the latter is a defensible technology.

1 The Puzzle

Kenya's urban logistics market simultaneously exhibits high demand for reliable goods movement, severe service fragmentation, thin formal margins, and the repeated failure of well-capitalized platforms attempting to formalize it. Conventional explanations—poor unit economics, premature geographic expansion, funding withdrawal, regulatory shock—describe proximate causes but do not identify the mechanism.

This paper proposes that the mechanism is **addressability**. In formally addressed markets, the cost of locating a destination is near zero and is embedded in a postal code. In informally addressed markets, the cost of location is borne **per drop**, in driver search time, telephone negotiation, landmark interpretation, and failed attempts. Critically, this cost is not a fixed overhead. It is **marginal, variable, and simultaneously suppresses the rate at which drops can be completed**—an effect that is multiplicative rather than additive.

2 Definitions

Definition 1 (Addressability). For drop j, addressability $a_j \in (0, 1]$ is the degree to which the destination can be located and accessed without verbal negotiation with the recipient.

- $a = 1$: geocoded, street-addressed, gate-identified, access point documented, recipient reachable
- $a \to 0$: landmark-only, mobile recipient, access negotiated at the point of delivery

Definition 2 (Address layer). A firm-level intervention raising effective addressability:

$$a_{\text{eff}} = a_N + (1 - a_N)\,\kappa\,L \tag{1}$$

where a_N is the national baseline, $L \in [0,1]$ is the firm's coverage of its drop base with verified address intelligence, and $\kappa \in [0,1]$ is the effectiveness of that intelligence in reducing search.

Definition 3 (Static and dynamic components). Addressability decomposes as:

$$a_j = a_j^S + a_j^D(n_j) \tag{2}$$

where a^S (geocode, landmark uniqueness, gate/floor specificity) is **purchasable in advance** via field verification, and a^D (recipient reachability, historical first-attempt rate) is **earned only through delivery volume** n_j. Illustrative weights: static 0.70, dynamic 0.30.

Definition 4 (First-attempt success). $p(a) = p_{\min} + (1 - p_{\min})\,a^\gamma$, increasing and concave in a, with $p_{\min} > 0$.

Definition 5 (Place ledger). A canonical record per **place** (not per customer) comprising an append-only event stream of every delivery attempt, from which all addressability, success, and service-time statistics are derived. Places are shared across customers; customer-specific data resides on a **delivery point** linking an account to a place and access point.

Definition 6 (Access point). A physical location within a place at which goods may be handed over—loading dock, reception desk, estate gate, locker bank—each with a capacity parameter.

3 The Rate Primitive

Expected service time per drop:

$$\mathbb{E}[\tau(a)] = \tau_{\text{load}} + \tau_{\text{travel}} + \sigma\,\frac{1-a}{a} + (1 - p(a))\,\tau_{\text{reattempt}} \tag{3}$$

where σ is the search-and-negotiation coefficient (minutes lost per drop to landmark interpretation and telephone negotiation), and $\tau_{\text{reattempt}}$ is the time cost of a failed delivery.

The drop rate:

$$R(a) = \frac{1}{\mathbb{E}[\tau(a)]} \tag{4}$$

Drops per trip (time-constrained):

$$N_{\text{trip}}(a) = \min\bigl(V_{\text{capacity}},\; H_{\text{trip}} \cdot R(a)\bigr) \tag{5}$$

This specification is important because **the binding constraint flips with addressability**. At high a, the vehicle fills before the shift ends (capacity binds). At low a, the shift ends before the vehicle fills (time binds). The second case characterizes informal-address markets, and it explains the observed underutilization of owned assets: capacity is purchased that cannot be filled within a working day.

Driver cost per drop:

$$c_{\text{driver}}(a) = w_{\text{driver}} \cdot \mathbb{E}[\tau(a)] = \frac{w_{\text{driver}}}{R(a)} \tag{6}$$

Labor cost per drop is inversely proportional to the drop rate. This is the central mechanism of the paper.

4 The Rate Equation

Net service fee per drop:

$$n_i = S \cdot T_i \cdot A(a_i) \cdot \max\Big[M,\ F + r_z\,\delta_i + k\,w_i\Big] \tag{7}$$

Client price:

$$P_i = (1 + v)\,n_i \tag{8}$$

with the addressing surcharge:

$$A(a) = 1 + \mu\,\frac{1-a}{a}, \qquad \mu < \lambda \tag{9}$$

Symbol	Meaning
S	Fuel surcharge multiplier, indexed to EPRA
T_i	Tier multiplier $= 1 + \pi + \sigma_t + \eta$
M	Minimum drop charge (floor)
F	Base pickup fee covering first D_0 km
r_z	Per-km rate by zone
δ_i	$\max(0, D_i - D_0)$
k	Per-kg rate by weight band
w_i	Chargeable weight (greater of actual and volumetric)
v	VAT rate
μ	Pass-through coefficient; $\mu < \lambda$

5 The Invariant

Per-drop contribution:

$$g(a) = \bar{n} \cdot p(a) \cdot A(a) - c_{\text{materials}} - \frac{w_{\text{driver}}}{R(a)} - (1 - p(a))\,c_r \tag{10}$$

The Invariant:

$$\boxed{\bar{n} \cdot p(a) \cdot A(a) > c_{\text{materials}} + \frac{w_{\text{driver}}}{R(a)} + (1 - p(a))\,c_r} \tag{11}$$

Read plainly: average net fee must exceed materials cost, plus **labor cost per drop (which is w_{driver} divided by the drop rate)**, plus expected reattempt cost.

Break-even drops per trip:

$$n^* = \frac{c_0}{\bar{n}\,p(a)\,A(a) - c_{\text{materials}} - \frac{w_{\text{driver}}}{R(a)} - (1 - p(a))\,c_r} \tag{12}$$

As a falls, the denominator shrinks **and** $N_{\text{trip}}(a) = H \cdot R(a)$ shrinks. The required number of drops rises while the achievable number falls. This is the multiplicative trap, expressed in one equation rather than two propositions.

6 The Model in Three Lines

(1) The rate. $R(a) = 1/\mathbb{E}[\tau(a)]$, where $\mathbb{E}[\tau(a)] = \tau_{\text{load}} + \tau_{\text{travel}} + \sigma(1-a)/a + (1-p(a))\tau_{\text{reattempt}}$.

(2) The price. $n_i = S \cdot T_i \cdot A(a_i) \cdot \max[M, F + r_z\delta_i + kw_i]$, and $P_i = (1+v)n_i$.

(3) The test. $\bar{n}\, p(a)\, A(a) > c_{\text{materials}} + w_{\text{driver}}/R(a) + (1-p(a))c_r$, equivalently $n^* = c_0/[\bar{n}p(a)A(a) - c_{\text{materials}} - w_{\text{driver}}/R(a) - (1-p(a))c_r] \leq N_{\text{trip}}(a)$.

Addressability enters only through line (1). Line (2) converts it into revenue. Line (3) decides whether the drop should happen at all. Every proposition in §7 is a statement about one of these three lines.

7 Propositions

Proposition 1 (Existence of a threshold)**.** *Under $p' > 0$, $A' < 0$, and $c'_{driver} < 0$, $g(a)$ is strictly increasing in a. If $g(0) < 0 < g(1)$, there exists a unique a^* such that $g(a^*) = 0$.*

Proposition 2 (The Rate Proposition)**.** *$R(a)$ is the sole intermediate variable. Contribution per drop, drops per trip, capacity per vehicle-hour, break-even volume, and SLA feasibility are all monotone in $R(a)$. Addressability enters the model **only** through $R(a)$ and through the variance of τ.*

Corollary 1 (The Core Problem)**.** *For any drop with $a_j < a^*$, contribution is negative **regardless of volume**. Scale does not resolve a per-drop failure.*

Proposition 3 (Schedulability)**.** *A guaranteed service window of width $\pm W$ is promisable only if $\text{Var}[\tau(a)] \leq \Psi(W)$ for some bound Ψ. Since $\text{Var}[\tau]$ is decreasing in a, there exists a level a_S below which **no window of commercially useful width is promisable at any price**. This is stronger than Proposition 1: below a_S the product is infeasible, not merely unprofitable.*

Proposition 4 (Drop value lowers the threshold)**.** $\partial a^*/\partial \bar{n} < 0$. *Higher-value drops are viable at lower addressability.*

Proposition 5 (Registry Proposition)**.** *The static component a^S is purchasable in advance of operations via field verification, and can lift a drop above a^* before any delivery occurs. However, $a^S \leq 0.70$ by construction; since schedulability (Proposition 4) depends principally on a^D, registration and operations are **complements, not substitutes**. Registration front-loads the foundation; operations completes it.*

Proposition 6 (Ledger Proposition)**.** *Effective addressability is not a stock but a cumulative function of delivery volume per place: $a_j(n_j)$ is increasing in n_j, with **zero marginal cost after the first delivery**. Consequently: (i) registry value is maximised by client selection for repeatability; (ii) place-level statistics accumulated across customers are a public good within the firm, so binding places to individual customers destroys value; (iii) the optimal estimator for a new delivery point at a verified place is a shrunk (place-prior) estimator, permitting a tight window to be quoted on the first delivery.*

Proposition 7 (Aggregation Proposition)**.** *An access point with capacity k converts up to k recipient drops into **one commercial drop**, multiplying revenue per stop and driving the presence-risk term $(1-p(a))c_r$ toward zero. However, an access point introduces a **fixed cost with no dispatch option**, so it is admissible only where the ledger already demonstrates demand above the utilization threshold q^*. This is formally identical to the asset-ownership constraint of Proposition 9: physical infrastructure sited in advance of demand reproduces the failure mode it is meant to solve.*

Proposition 8 (Asset ownership raises the threshold). *Ownership converts c_0 from variable to fixed. Break-even volume $N^* = f/(\bar{n}p(a)A(a) - c_1(a))$ then requires volume that $N_{trip}(a)$ may be unable to supply as a falls. Asset-light capacity preserves the option not to dispatch.*

Proposition 9 (Risk Proposition). *In high-value B2C, expected loss enters the invariant as a first-order term:*

$$\mathbb{E}[loss] = (\rho_{fraud} + \rho_{theft} + \rho_{damage})\, V \tag{13}$$

and the gate condition becomes:

$$\left[\bar{n}\, p(a)\, A(a) - c_1(a)\right] > \mathbb{E}[loss] \tag{14}$$

$\mathbb{E}[loss]$ *is a function of* ***recipient history*** *rather than place addressability, so the firm must accumulate* ***two*** *ledgers: place and recipient. The recipient identifier (a persistent phone-linked ID) is portable across places, which is what permits underwriting a consumer at a new address.*

8 The Core Problem, Stated

In low-addressability markets, the drop rate is suppressed by search time and failed attempts. Because labor cost per drop is the reciprocal of the drop rate, and because capacity per vehicle-hour is proportional to it, low addressability simultaneously raises unit cost and lowers the revenue ceiling. Break-even drops per trip therefore rises faster than any feasible price premium can offset. Viability requires raising either drop value or effective addressability above a critical threshold. Mass-market strategies in informal-address economies raise neither and are structurally loss-making regardless of capital, scale, or execution quality.

9 Calibration and Reconciliation

Illustrative parameters: $\bar{n} = 800$ (mass-market standard drop), $p(a) = 0.6 + 0.4a$, $c_{\text{materials}} = 100$, $w_{\text{driver}} = 250$, $c_r = 200$, $c_0 = 2{,}500$. *(All values to be replaced with field data.)*

Case A — mass-market, standard tier ($\bar{n} = 800$):

- $g(0.5) \approx +250 \rightarrow$ viable
- $g(0.3) \approx -163 \rightarrow$ fails
- $a^* \approx 0.36$

Informal urban micro-retail destinations sit at approximately $a \approx 0.2$–0.3. **The mass market lay below the threshold.**

Case B — B2B contract ($\bar{n} = 2{,}000$):

- $g(0.2) \approx +196 \rightarrow$ viable
- $a^* \approx 0.17$

A 2.5× increase in drop value lowers the threshold below the informal baseline. **This is the formal reason the B2B contract segment is viable where the mass market is not.**

Case C — Registry lift ($a^S = 0.70$ at a provisional place, $a_N = 0.25$):

$$a_{\text{eff}} = 0.25 + 0.75 \times 0.70 = 0.775 \tag{15}$$

clears $a^* = 0.36$ at standard tier with headroom, before a single delivery.

Case D — Aggregation (locker bank, $k = 40$ at a verified place):

- One commercial stop, 40 parcels, presence risk eliminated
- Revenue per vehicle-hour multiplied $\sim k$-fold at the stop level

Reconciliation with observed failures.

- **Sendy** operated at low $\bar{n}$, low a, with fulfilment expansion raising c_0 and a retroactive principal-VAT ruling raising effective cost by 16% of gross—pushing g below zero across the book.
- **Twiga** owned c_0 (warehouses, fleet, farming), invoking Proposition 9, while operating in fresh produce with low $\bar{n}$ and high variance. It required N_{trip} above a level that informal addressing made unreachable.
- Neither invested in raising a through either the registry route (Proposition 6) or the ledger route (Proposition 7). Both solved for volume and price. **The binding constraint was neither.**

10 Testable Predictions

1. Contribution per drop is **bimodal**, not continuous, with a gap around a^*.
2. Drops per vehicle-hour is the **strongest single predictor** of corridor profitability, dominating average fee per drop.
3. Firms with a registry and ledger show **higher drops per vehicle-hour at equal route length** than firms without.
4. Asset-owning firms show **fatter left tails** in contribution per drop.
5. Failed-delivery rates are predicted by **landmark-only addressing and recipient unreachability**, not by distance.
6. Places with $n_j \geq 10$ support **materially tighter quotable windows** than places with $n_j < 3$.
7. Access points demonstrate **parcels per door per day** that either clears or fails a threshold within 90 days; the distribution should be bimodal, not uniform.

11 Scope and Limitations

The paper proves a **conditional** proposition: given a^* and the firm's ability to hold a or $\bar{n}$ above it, viability follows. It does not prove any specific firm will succeed. Parameter estimates for σ, κ, γ, $p_{\min}$, and the static/dynamic split of a require field calibration. The model assumes stable partner costs and no principal-tax reclassification shock, which should be modelled as an increase in effective cost rather than as a change in a. Proposition 10 treats $\mathbb{E}[\text{loss}]$ as exogenous to the delivery design; in practice security controls endogenise it, and the control-effectiveness function requires separate estimation.

12 Conclusion

Addressability is not a background condition of the operating environment. It is the decisive variable in the unit economics of last-mile logistics, and it operates through a single channel—the drop rate. Firms that treat addressability as fixed inherit its consequences. Firms that

treat it as a design problem build two things: a **registry** that raises the static component of addressability in advance of operations, and a **ledger** that accumulates the dynamic component as a byproduct of revenue. The registry front-loads the foundation; the ledger compounds it. That is the theoretical basis for the firm described in the accompanying white paper.